

Project Design Phase-II
Technology Stack (Architecture & Stack)

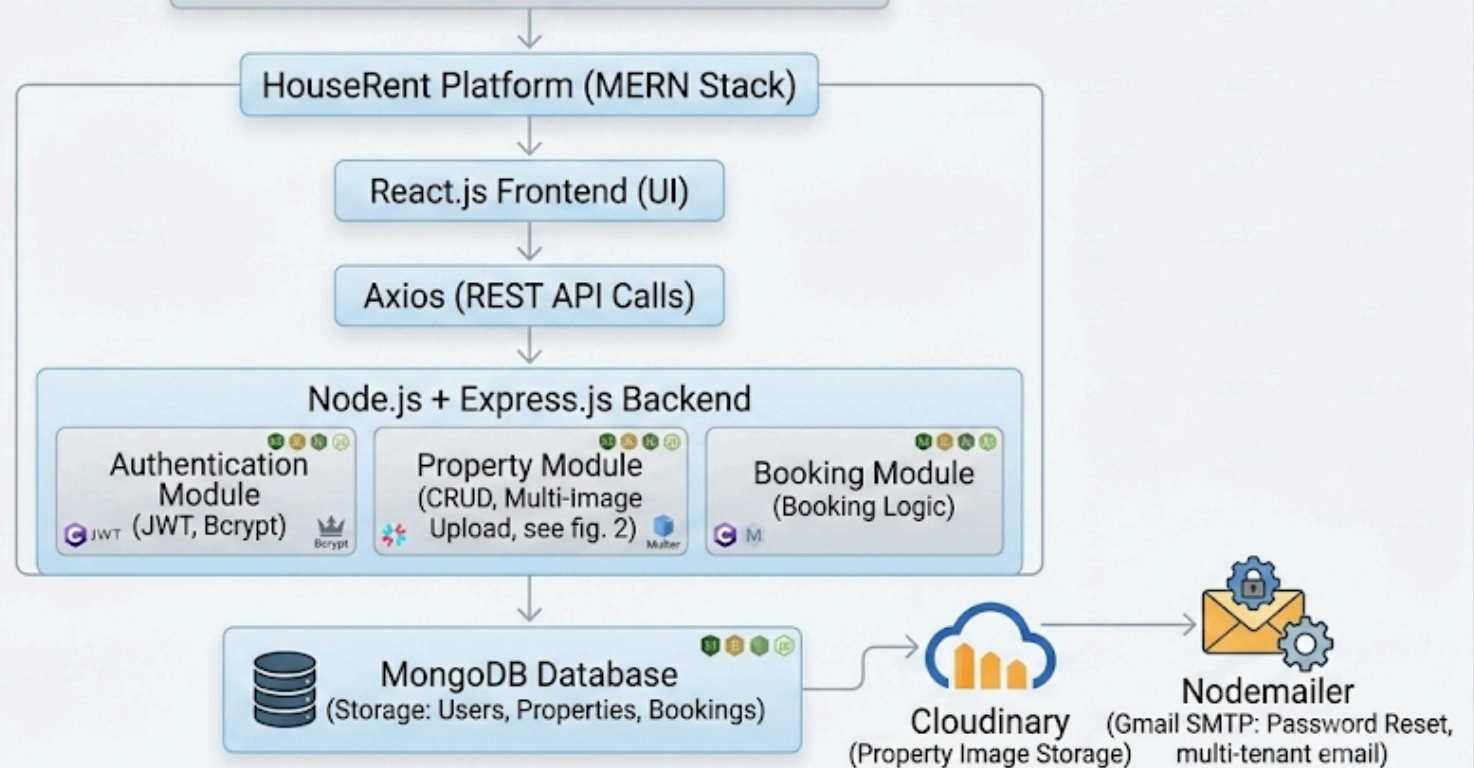
Date	1 July 2026
Team ID	
Project Name	
Maximum Marks	4 Marks

Technical Architecture – HouseRent (MERN Stack House Rental Platform)

Architecture Diagram

Figure 1: High-Level Platform Architecture

User / Owner / Administrator



Comprehensive MERN Stack Platform Architecture and Data Flow

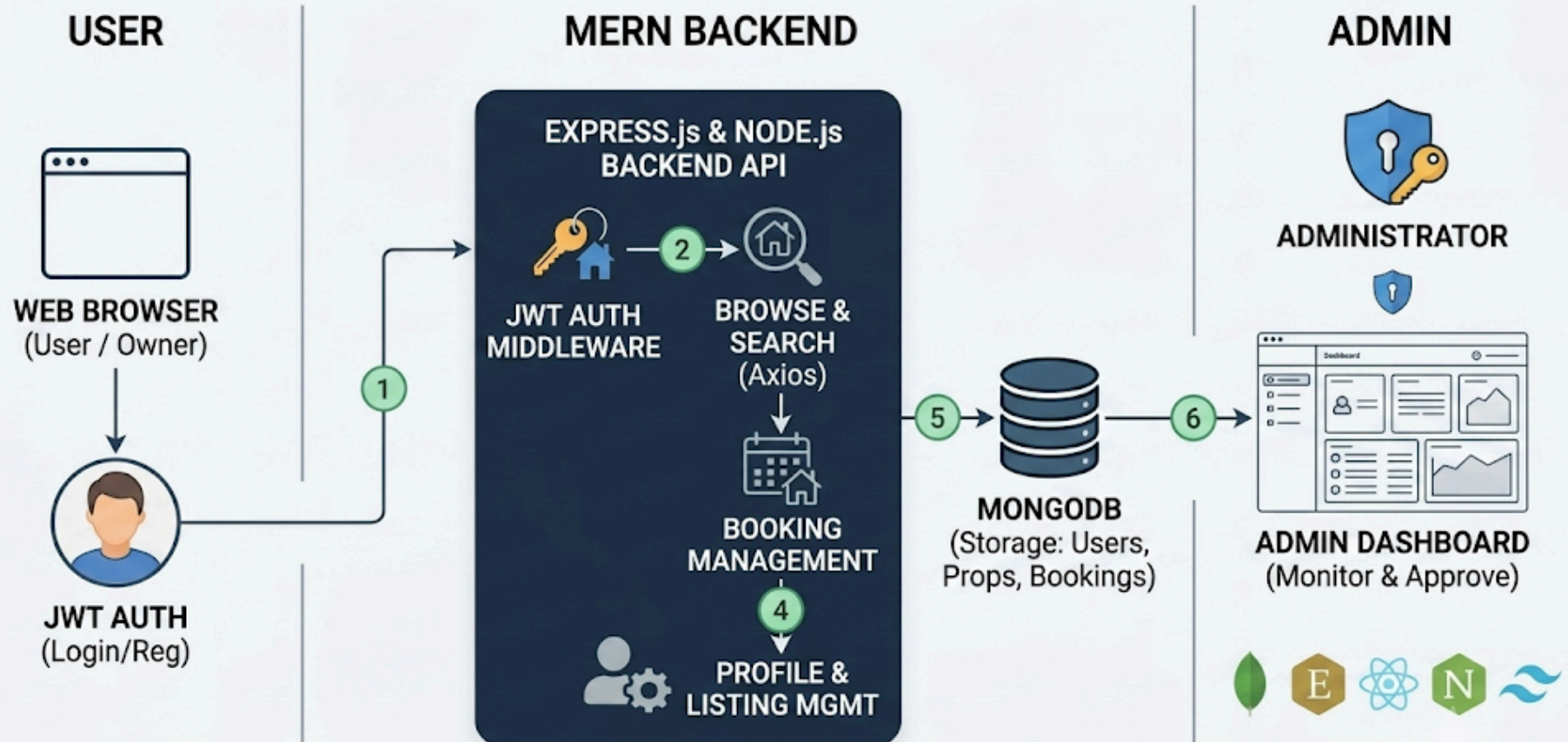


Figure 1: Architecture and data flow of the HouseRental platform

Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for Users, Owners, and Admin	React.js, HTML5, CSS3, Tailwind CSS, JavaScript
2	Application Logic-1	User authentication, authorization, and business logic	Node.js, Express.js
3	Application Logic-2	Property management (Add, Edit, Delete, View)	Express.js, Mongoose
4	Application Logic-3	Booking management, Owner approval, Admin operations	Node.js, Express.js
5	Database	Stores Users, Properties, and Bookings	MongoDB, Mongoose
6	Cloud Database	Cloud-hosted MongoDB database	MongoDB Atlas
7	File Storage	Stores uploaded property images	Cloudinary, Multer

8	External API-1	Image upload and storage service	Cloudinary API
9	External API-2	Sends password reset emails	Gmail SMTP (Nodemailer)
10	Machine Learning Model	Not Applicable (Current Version)	—
11	Infrastructure (Server/Cloud)	Application deployment on local system or cloud	Local Server / Render / Vercel / MongoDB Atlas

Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Frontend and backend frameworks used for application development	React.js, Node.js, Express.js, MongoDB, Tailwind CSS
2	Security Implementations	Secure authentication, password encryption, role-based access control, protected routes	JWT, Bcrypt.js, HTTPS, Role-Based Access Control (RBAC)

3	Scalable Architecture	Three-tier architecture with separate frontend, backend, and database layers for easy scalability	MERN Stack (React, Express, Node.js, MongoDB)
4	Availability	Cloud-based database and image storage ensure high availability and reliable access	MongoDB Atlas, Cloudinary
5	Performance	Fast API communication, optimized database queries, efficient image delivery, and responsive UI	Axios, Mongoose, MongoDB Indexing, Cloudinary CDN, React.js